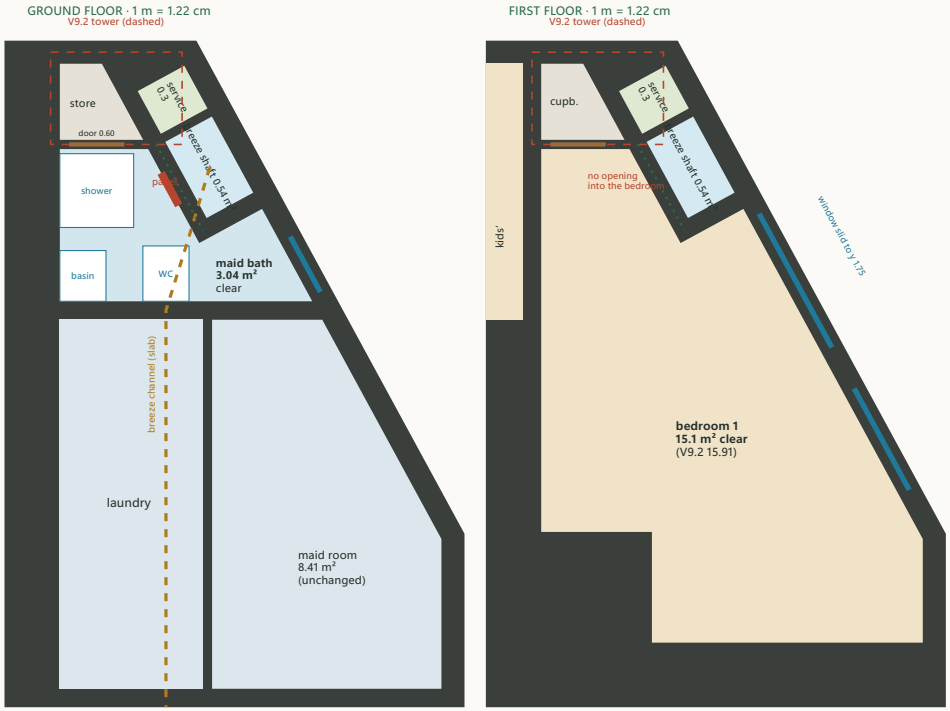
Malqaf tower study · a slim shaft along the cut wall, a tall stone tower set back on the roof · Taif / Palm House

SUPERSEDED BY L5 (V10.0, owner, 9 Nov 2026: `studies/malqaf_L5.typ`). Was adopted at V9.3: L1 (the north-east cut corner), head (c), 3.5 m above the attic coping, the flat Hijazi cornice · the slim shaft along the cut wall with the store and the cupboard, the HRV in the plant room behind the attic (moved 0.5 m south, y 2.9–4.3, so the roof hatch in the tower's south face is reached), the breeze channel at its full width; applied in the design files. This study stays the record of the comparison.

STUDY ONLY (8 Nov 2026): the design files are not changed; the owner chooses after this study. Owner's intent: a real, elegant malqaf, the symbol of the house, taller and slimmer, in honey stone, cream plaster and black steel with simple cornices; no awkward corners inside. All sizes assumed: the MEP and structural engineers confirm.



1 · The shaft today (V9.2)

- A rectangle at the tip of the cut corner, **1.2 m wide × 0.75 m deep clear = 0.9 m²** (1.43 m² on the wall centre lines; the breeze shaft 0.54, the service compartment 0.28); the slab openings ≈ 0.98 m². Its east wall leaves a **wedge** against the cut wall in the maid bath (the shelf niche) and in bedroom 1 (the cupboard): the awkward corners the owner objects to. Above the roof it rises from the 5.84 m² HRV base: the "box on a box".

2 · The proposal: a slim shaft hugging the cut wall

- **Clear 0.57 m deep × 1.57 m long along the cut = 0.9 m²**, between the cut wall's inner face and **one new 200 mm wall parallel to the cut**, closed by a short end wall square to the cut; the tip beyond it a **solid corner pier** (it stiffens the corner). Split by an airtight partition (0.10): the **breeze shaft 0.54 m²** (south part) and the **service compartment 0.3 m²** (north part: the HRV's two Ø250 supply / extract risers to the ground floor; the HRV's intake is now taken at the tower's head, § 3).
- **The rooms (straight walls):** north of the line y 1.0 a 100 mm partition closes a **store** off the maid bath (0.56 m² clear, door 0.60) and, upstairs, a **cupboard** in bedroom 1 (0.56 m²); south of it each room gets **one straight wall parallel to the cut** (1.06 m, dotted green) and a square return; its corners 90°, 118.5° or, where the cut meets a cross wall, the cut's own 61.5° (8 Nov 2026 correction: an earlier angle test missed it).

Room	axes m²	clear m²	V9.2 clear
Maid bath (G05)	3.9	**3.04**	3.85
Its store (new)	0.87	0.56	–
Maid room (G06)	9.55	8.41	8.41
Bedroom 1 (F11)	16.97	**15.1**	15.91
Its cupboard (new)	0.87	0.56	–
Shaft (axes / clear)	1.59	0.9	0.9

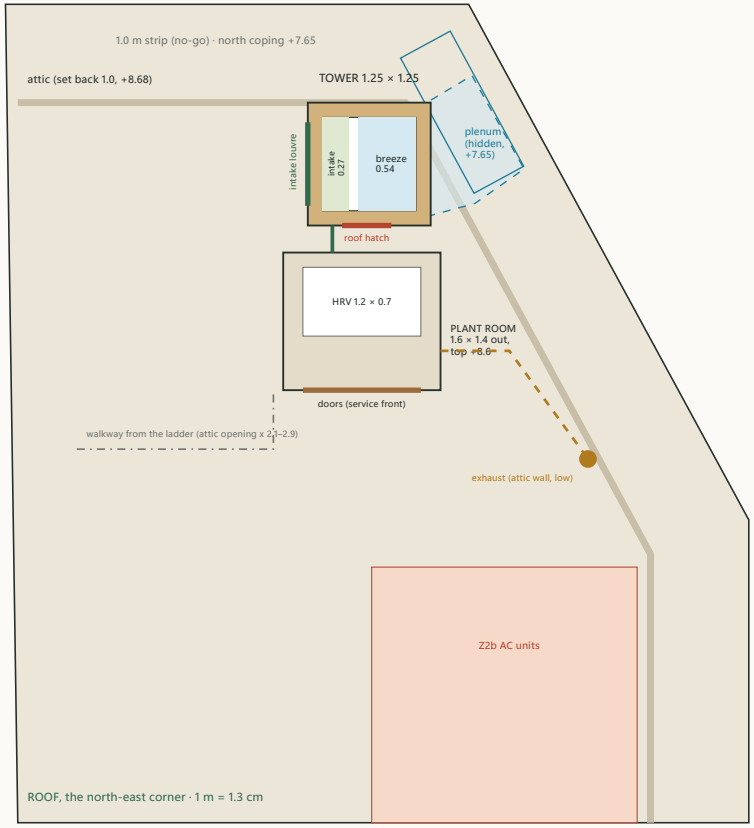
- **The price of the straight walls:** the maid bath −0.81 m² and bedroom 1 −0.81 m² clear, against a usable store and cupboard of 0.56 m² each (net −0.25 / −0.25). **Option:** if the ground floor's HRV risers go down the west service shaft instead, the slim shaft is the breeze shaft alone (≈ 0.94 m long) and both rooms win back ≈ 0.35 m².

3 · Cleaning without entry

- **From the roof:** the head's bird mesh in slide-down cassettes, one per quarter, lowered on cords to a **small roof hatch** (0.50 × 0.50) in the tower's south face inside the attic; the **top damper** just above the plenum, reached through the same hatch; the plenum's lid is a second hatch. **From the ground floor:** one **access panel 0.40 × 0.40** (red) at the breeze shaft's foot, opening from the **maid bath** (sill +0.10, gasketed, screw-fixed): the sump, the channel's mouth. **No opening into bedroom 1.** The shaft's inside **smooth-rendered** (sealed), so dust does not cling.

Study checks **passed** (14) · the clear areas: the walls' half thicknesses edge by edge (0.125 outer, 0.10 for 200 mm, 0.05 for the store partition), the same allowances as the presentation's schedule · the bath's fixtures indicative (the interior study re-plans them) · bedroom 1's first cut window slides 0.15 m south (y 1.75) to clear the end wall.

Above the roof · the HRV plant room behind the attic, the hidden plenum, the tower set back on the attic's corner



4 · The HRV unit off the tower: a plant room hidden behind the attic

- **Where and how big:** inside the attic, south of the tower, x 8.7–10.3, y 2.4–3.8 (1.6 × 1.4 m outside, 2.24 m²), light insulated walls, a slab at +8.6: **below the attic coping (+8.68), so hidden from the street, the lane and the courtyard**; ≈ 1.15 m clear inside: a service compartment, not a room (the 1.2 × 0.7 × 0.7 unit on a 0.10 plinth, its front at the doors).
- **Service access:** a pair of doors on its south face (filters and the core pulled out), reached on the roof walkway from the maintenance ladder through the attic opening (the strip in front of the attic stays a no-go zone).
- **Air:** the **intake** from the tower's own intake compartment (0.27 m²; its louvre on the tower's west face just below the band, high above the roof: the cleanest air), dropping to the roof and through a 0.3 m duct into the plant room; the **exhaust** led along the attic to a low grille in the attic's cut wall (+7.7), blowing east over the lane side, hidden behind the coping. Separations: intake ↔ exhaust 4.35 m or more (ASHRAE 62.1 ≥ 3 m), the exhaust 1.1 m from the condensers (≥ 1 m). The first floor fed straight through the roof slab; the ground floor through the shaft's service compartment.

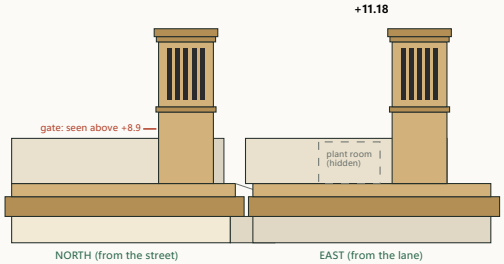
5 · The tower above the roof: slim, square, set back

- **Section 1.25 × 1.25 m outside** (0.15 walls, 0.95 × 0.95 clear = 0.9 m²: the breeze part 0.54, the partition, the intake 0.27), square to the house like the attic and the courtyard. **Set back:** its north face on the attic's north line (1.0 m behind the north facade), its west face over the wall W3 (x 8.95), its north-east corner 0.77 m behind the cut facade: it reads as **the attic's corner rising**, not a box perched on the facade's edge. The attic's coping line wraps round it as its plinth.
- **The transition hidden:** the slim shaft rises inside the cut wall to the roof; a low **plenum** (the roof slab omitted over it, +7.00 … +7.65, smooth-rendered, partitioned) turns the breeze from the shaft's top into the tower's foot, **below the north coping (+7.65): never seen**. Its two gentle bends cost little (included in § 7).
- **Bird mesh and damper as decided (air v3.1):** a wide 20 mm bird mesh only, the + divider and the back-draught flaps in the head, the motorised spring-return damper at the tower's foot (the roof hatch).
- **Structure (note for the engineer):** the tower stands on the roof slab over the W3 line and the new 100 / 200 wall y 1.0 (to be RC under the tower), with an RC ring beam at the roof and the plenum box tying it to the shaft's walls; self-weight 55.2 – 71.2 kN; the wind (assumed q 1.1 kN/m², Cf 1.3) gives 6.9 – 8.7 kN and 13.5 – 21.3 kN·m at the roof: modest, but the slender variants need starter bars from the slab and a check of the slab's local bending. To confirm with SBC 301 wind data.

The variants · north and east elevations (the roof zone, 1 m = 0.58 cm) · heights above the attic coping +8.68

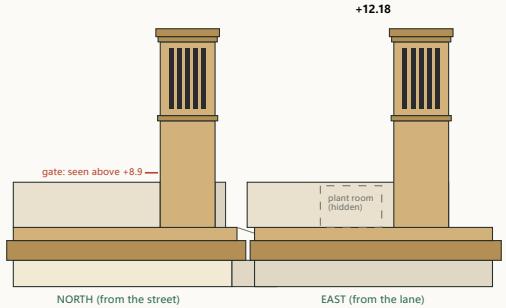
References (owner): the Yazd badgirs, e.g. the “father’s home” windcatcher in Ashkezar: a tall slim tower with a plain stone shaft, a cornice band, a head with many slender vertical openings (ribs) on each face and a simple crowning cornice. Heads: **(a)** one semicircular (Roman) arch per face; **(b)** the Yazd type, 5 slender slots 0.11 m wide per face between ribs; **(c)** 4 slots 0.14 m wide, each with a small semicircular top, linking them to the house’s arches (the courtyard arcade, the enclosure windows). Tops: a flat Hijazi cornice (two plain steps) or a small dome. The red tick: what is seen from the gate begins above it.

(b) Yazd slots · 2.5 m · flat



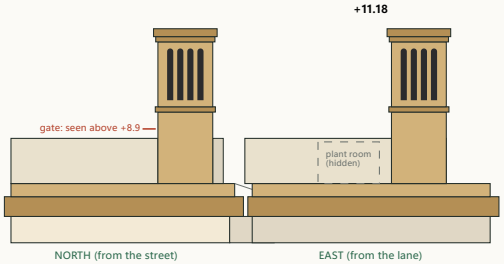
top +11.18 (11.48 m above the ground) · 3.88 m above the roof, proportion 3.1 : 1 (2 : 1 above the attic) · 5 openings per face, 0.65 m² free · seen from the gate 2.28 m · from the courtyard its top 0.04 m

(b) Yazd slots · 3.5 m · flat



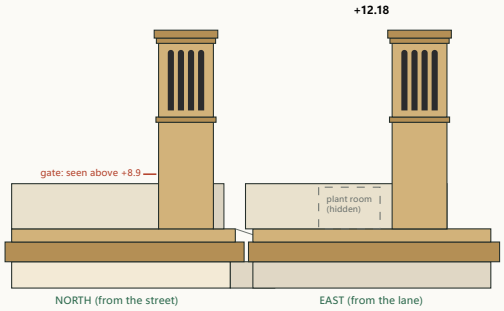
top +12.18 (12.48 m above the ground) · 4.88 m above the roof, proportion 3.9 : 1 (2.8 : 1 above the attic) · 5 openings per face, 0.76 m² free · seen from the gate 3.28 m · from the courtyard its top 1.04 m

(c) arched slots · 2.5 m · flat



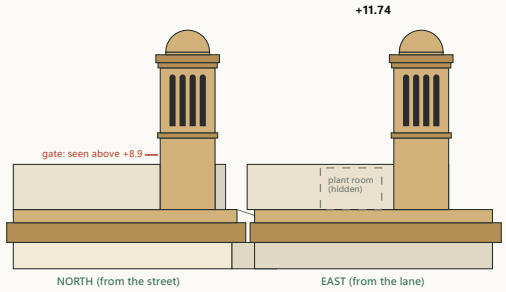
top +11.18 (11.48 m above the ground) · 3.88 m above the roof, proportion 3.1 : 1 (2 : 1 above the attic) · 4 openings per face, 0.65 m² free · seen from the gate 2.28 m · from the courtyard its top 0.04 m

(c) arched slots · 3.5 m · flat



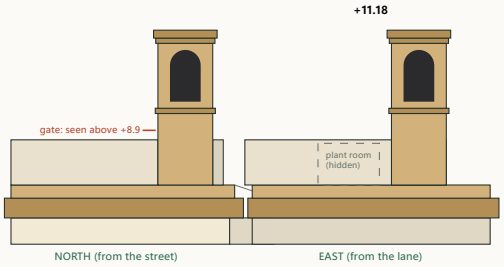
top +12.18 (12.48 m above the ground) · 4.88 m above the roof, proportion 3.9 : 1 (2.8 : 1 above the attic) · 4 openings per face, 0.76 m² free · seen from the gate 3.28 m · from the courtyard its top 1.04 m

(c) arched slots · 2.5 m · dome



top +11.74 (12.04 m above the ground) · 4.44 m above the roof, proportion 3.6 : 1 (2.4 : 1 above the attic) · 4 openings per face, 0.65 m² free · seen from the gate 2.84 m · from the courtyard its top 0.6 m

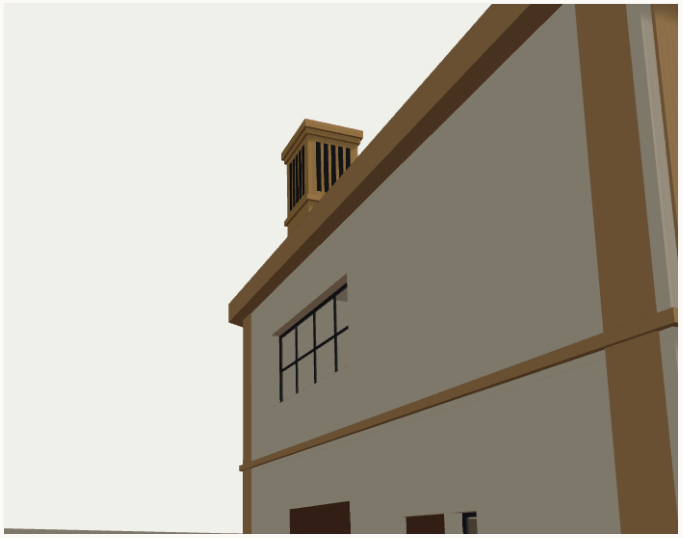
(a) one Roman arch · 2.5 m · flat



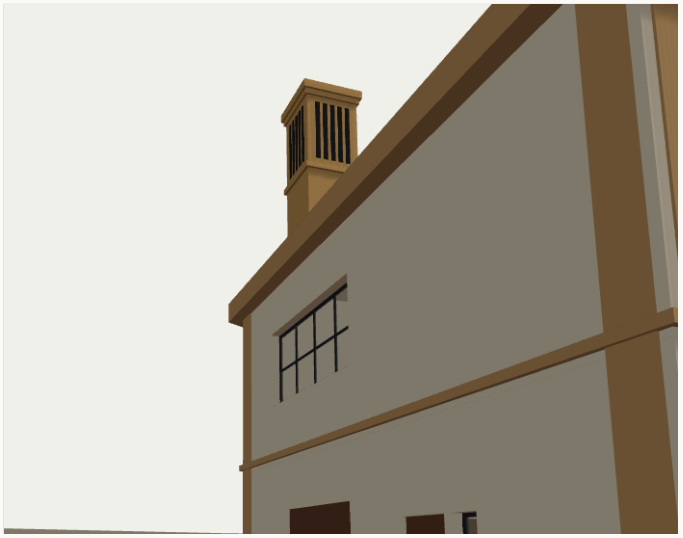
top +11.18 (11.48 m above the ground) · 3.88 m above the roof, proportion 3.1 : 1 (2 : 1 above the attic) · 1 opening per face, 0.77 m² free · seen from the gate 2.28 m · from the courtyard its top 0.04 m

From the gate · the same camera as char_gate.png (the north facade from just inside the farm's main gate)

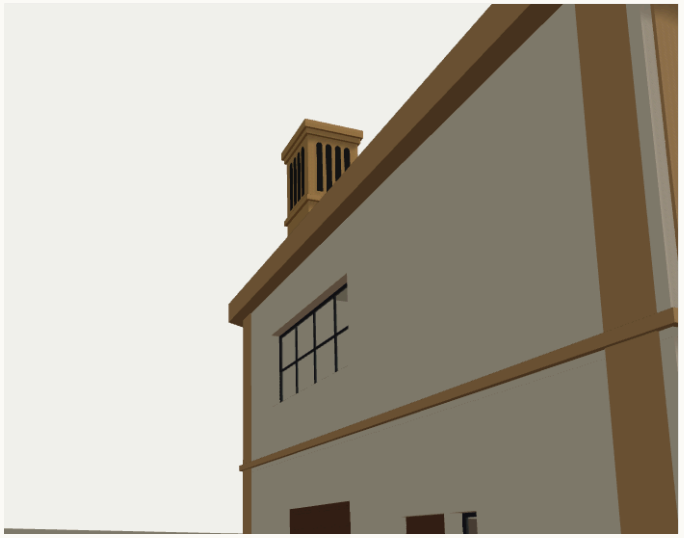
3D: `3d/malqaf_variants.py` (the current model with the present head and its HRV base taken out, the roofline rebuilt, the variant added; the plant room and the plenum built but hidden behind the attic and the coping). Style study, provisional; the stone colours of the character palette.



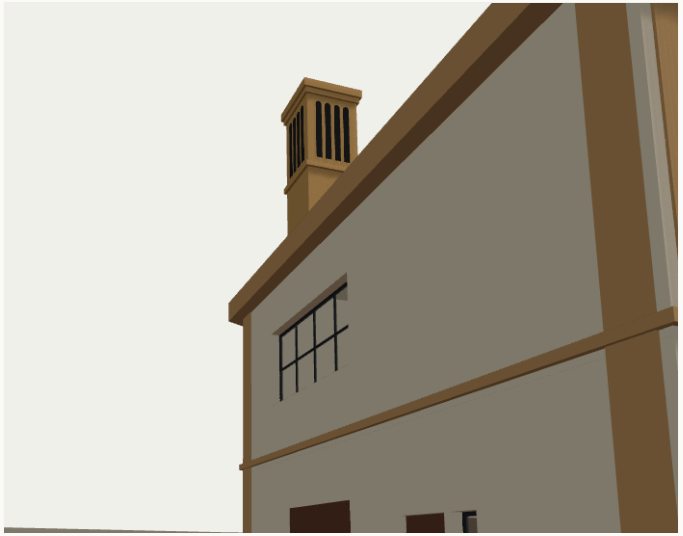
(b) Yazd slots · 2.5 m · flat · top +11.18



(b) Yazd slots · 3.5 m · flat · top +12.18



(c) arched slots · 2.5 m · flat · top +11.18



(c) arched slots · 3.5 m · flat · top +12.18



(c) arched slots · 2.5 m · dome · top +11.74



(a) one Roman arch · 2.5 m · flat · top +11.18

The comparison and the recommendation

Variant	Top	Prop.	Free / face	Breeze	Seen: gate	Seen: court	Weight	Wind M	Cost	Reading
(b) Yazd slots · 2.5 m · flat	+11.18	3.1 : 1	0.65 m²	1.02	2.28 m	0.04 m	57 kN	13.5 kN·m	1.2	true Yazd language; compact: a lantern on the attic
(b) Yazd slots · 3.5 m · flat	+12.18	3.9 : 1	0.76 m²	1.04	3.28 m	1.04 m	71.2 kN	21.3 kN·m	1.42	the closest to the Ashkezar reference: a real tower; the many dark slots read strongly
(c) arched slots · 2.5 m · flat	+11.18	3.1 : 1	0.65 m²	1.02	2.28 m	0.04 m	56.9 kN	13.5 kN·m	1.3	the house's arches in miniature; a little short for the slots
(c) arched slots · 3.5 m · flat	+12.18	3.9 : 1	0.76 m²	1.04	3.28 m	1.04 m	71.2 kN	21.3 kN·m	1.53	**slim, tall, tied to the house's arches: the symbol of the house**
(c) arched slots · 2.5 m · dome	+11.74	3.6 : 1	0.65 m²	1.02	2.84 m	0.6 m	59.9 kN	17.6 kN·m	1.45	the dome reads as a minaret / a shrine: foreign to a Hijazi house
(a) one Roman arch · 2.5 m · flat	+11.18	3.1 : 1	0.77 m²	1.02	2.28 m	0.04 m	55.2 kN	13.5 kN·m	1	one arch per face: a lookout or a belfry, not a windcatcher

Top: the highest point (the dome's crown). **Prop.:** the height above the roof finish to the width. **Free / face:** the openings of one face (the bird mesh lets 90 % through). **Breeze:** the courtyard breeze in the same wind relative to the present head (1.00): the head's higher, freer wind (a power law 0.20 with height) and its entry loss in the air study's loss chain (the 0.10 m² throat under the maid bath dominates, so the head changes the breeze only by a few per cent); the stack in still air grows with the height about as much. **Seen:** the visible height from the gate (above +8.9, over the north coping) and from the courtyard's south edge (above +11.14, over the roof's edge). **Weight:** the tower above the roof finish. **Wind M:** the overturning moment at the roof (assumed wind). **Cost:** relative to (a) at 2.5 m (stone, the ribs and the arched tops, the height); the plant room and the plenum are common to all.

6 · Recommendation

- **Inside: the slim shaft along the cut wall** (§ 2): the walls of the maid bath and bedroom 1 straight, one wall parallel to the cut, the tip a solid pier, the store and the cupboard north of y 1.0; cleaned from the roof and from one panel in the maid bath; the HRV unit in its plant room behind the attic.
- **Above the roof: (c) at 3.5 m above the attic coping with the flat Hijazi cornice** (top +12.18): slim (3.9 : 1 above the roof), set back on the attic's corner, a plain honey-stone shaft, a cornice band, 4 slender openings per face each with a small semicircular top (the house's arches in small), the bird mesh behind them, a two-step crowning cornice. It is the one that reads as **a real malqaf and the symbol of the house**: from the gate 3.28 m of it stands above the coping, and from the courtyard its crown just shows over the north roof (1.04 m). Breeze ≈ 1.04 of today's; cost ≈ 1.53 × the simplest.
- **If the owner prefers the pure Yazd language:** (b) at 3.5 m, the same tower with straight slots. **If a lower tower is wanted:** (c) at 2.5 m. **Not recommended:** the dome (it reads as a minaret or a shrine, foreign to a Hijazi house) and the single arch (a lookout, not a windcatcher; its one big opening per face is the least "malqaf").
- **Next:** the owner chooses; then the design files (the shaft and the rooms in the CSVs and the studies, the plant room, the roof, the structure, the 3D) in one work round, the MEP engineer's check of the breeze and the HRV ducts, the structural engineer's of the tower on the slab.

ملخص

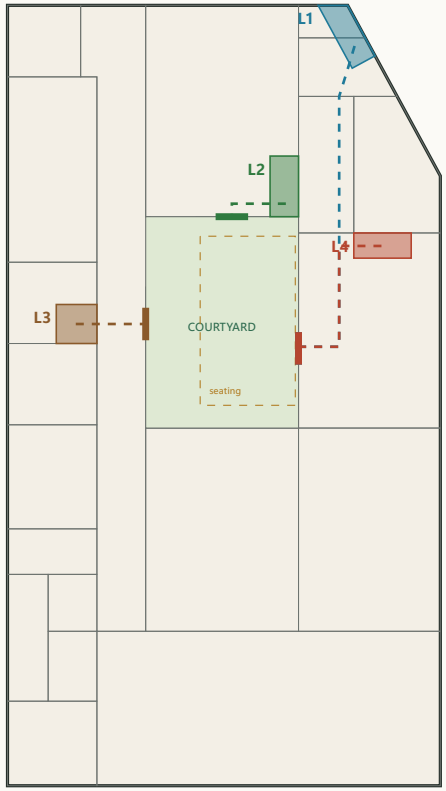
في الداخل: مجرى رفيع بعمق 0.575 م يحاذي جدار القطع (≈ 0.90 م² صافيا: مجرى النسيم 0.54 م² وحجرة الخدمة 0.30 م² بينهما فاصل محكم)، فيصبح لكل من حمام الخادمة وغرفة النوم 1 جدار مستقيم واحد موازٍ للقطع بلا زوايا غريبة، ويُستفاد من الطرف الشمالي مخزنا للحمام ودولابا لغرفة النوم. التنظيف دون دخول: من السطح (شبك الطيور على أدراج تُنزَل إلى فتحة صغيرة، والمخمد العلوي من الفتحة نفسها) ومن فتحة صيانة 0.40 × 0.40 م عند قاعدة المجرى من حمام الخادمة، ولا فتحة في غرفة النوم.

فوق السطح: تُنقل وحدة التهوية إلى غرفة معدات صغيرة خلف العلّية لا تُرى، ويرتفع البرج نحيلا مربعا 1.25 × 1.25 م من الحجر العسلي، مرتدا عن حافة الواجهة على زاوية العلّية. **التوصية:** الرأس (ج) بفتحات رأسية نحيلة لكل منها قوس نصف دائري صغير، بارتفاع 3.5 م فوق غطاء العلّية مع كورنيش حجازي مسطح: ملقف حقيقي ورمز للبيت. البديل: (ب) فتحات يزدية مستقيمة. لا يُنصح بالقبة ولا بالقوس الواحد.

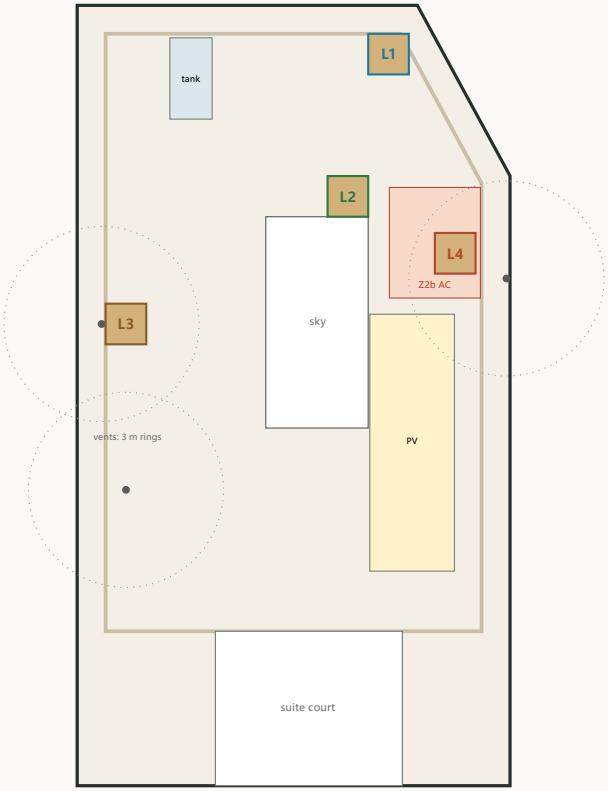
Sources: studies/air_system.typ (the present tower, its sections and the breeze loss chain), studies/roof.typ (the attic, the copings, the sight lines), data/rooms_gf.csv / rooms_ff.csv (V9.2), 3d/character.yaml (the gate camera). Concept only: every size assumed; nothing here is for construction.

Where the tower stands · four positions compared (the same slim shaft, the recommended tower (c) at 3.5 m, top +12.18)

STUDY ONLY, the design files unchanged. Looked at and dropped before the four: **the courtyard's kitchen side** (the kitchen's west wall carries the sink window, the water + tree enclosure and the outdoor TV; upstairs the suite lobby is only 1.35 m wide) and **the riwaq side** (the family's circulation under the gallery). Prevailing wind: north-west. Both key plans: 1 m = 0.43 cm.



GROUND FLOOR · shafts, breeze routes (dashed), grilles (bold)



ROOF · towers 1.25 × 1.25, attic, Z2b, vents

L1 · the north-east cut corner (today; pages 1–5)

Breeze: 10.6 m from the shaft's foot to the grille (through the 0.10 m² throat under the maid bath), ≈ 0.99 × today's flow. **Intake:** 8.3 m to the nearest vent (≥ 3), 4.6 m to the condensers (≥ 2). **Rooms:** GF maid bath −0.81 m² (+ a store 0.56) · FF bedroom 1 −0.81 m² (+ a cupboard 0.56). **Walls:** straight: one wall parallel to the cut per room + a square return. **Structure:** on the roof slab over W3 and a new RC wall y 1.0; a plenum box ties it to the shaft (offset 0.3–1.0 m). **Roof:** the attic's NE corner; the plant room behind it; Z2b, the tank, the PV untouched. **Seen:** from the gate 3.28 m, from the street 4.09 m, from the courtyard 1.04 m. **Cost:** ≈ 1.

L2 · the courtyard's north-east corner (the women's majlis's SE corner)

Breeze: 2.1 m from the shaft's foot to the grille, ≈ 1.56 × today's flow. **Intake:** 6.3 m to the nearest vent (≥ 3), 2.7 m to the condensers (≥ 2). **Rooms:** GF women's majlis −1.37 m² (a nib in its SE corner; its courtyard glass ends 0.1 m sooner) · FF family living −1.37 m² (a nib in its SE corner by the courtyard glass). **Walls:** straight: a rectangular nib against W3 in each room. **Structure: the best:** on the crossing of W3 (x 8.95) and the courtyard's north wall line (y 6.5), straight over its shaft, no plenum, no transfer. **Roof:** the sky hole's NE corner, inside the attic; the guard meets the tower; Z2b 1.9 m east (an intake ≥ 2 m away). **Seen:** from the gate 0 m, from the street 1.42 m, from the courtyard 4.88 m. **Cost:** ≈ 0.85.

L3 · the west band, beside the west service shaft (the utility room)

Breeze: 2.2 m from the shaft's foot to the grille, ≈ 1.53 × today's flow. **Intake:** 1.8 m to the nearest vent (≥ 3), 9 m to the condensers (≥ 2). **Rooms:** GF utility room −1.2 m² (5.81 → ≈ 4.6) · FF **bedroom 2's en-suite −1.2 m²** (5.85 → ≈ 4.4 on axes): its fixtures stand on that very service wall. **Walls:** straight (the bay between the west shaft, the riwaq wall and the utility's south wall). **Structure:** on the west outer wall and the shaft's walls: good. **Roof:** the west strip; **the west stack's vent beside the intake: re-route it ≥ 3 m away**. **Seen:** from the gate 2.95 m, from the street 0 m, from the courtyard 0 m. **Cost:** ≈ 0.95.

L4 · over the kitchen, against its north wall (the east band)

Breeze: 5.7 m from the shaft's foot to the grille, ≈ 1.51 × today's flow. **Intake:** 3 m to the nearest vent (≥ 3), 1.9 m to the condensers (≥ 2). **Rooms:** GF kitchen −1.4 m² **and the route from the laundry to the service door closed (0.3 m left)** · FF **a box standing in the middle of the master suite's bath**. **Walls:** straight in the kitchen; not in the suite bath. **Structure:** the kitchen's north wall is a non-load-bearing partition: a new RC wall and footing. **Roof: inside Z2b (the condensers move) and 3 m from the east vent**; the future PV. **Seen:** from the gate 0 m, from the street 0.58 m, from the courtyard 0 m. **Cost:** ≈ 1.15.

Location checks **passed** (4) · **Breeze:** the air study's loss chain (the head, the flaps, the damper, the bend and the plenum, the grille, the channel's friction by its length, the throat where there is one, the extra plenum where the tower is offset), the same head and wind for all: relative flows. **Seen:** the height of the tower (top +12.18) above the sight line from the gate (the char_gate camera), from the north street 14 m off the facade and from the courtyard's south edge, over the copings, the attic and the courtyard's roof edges. **Cost:** relative to L1 (the shaft, the channel, the plenum, the structure, the knock-on moves), the tower itself the same.

The location comparison and the recommendation

Position	Breeze route	Breeze	Intake	Rooms / walls	Structure	Roof	Gate · street	Courtyard	Cost
L1 · NE cut corner	10.6 m	◐ 0.99 (1.44 without the throat)	● 8.3 / 4.6 m	● service rooms	◐	●	● 3.3 · 4.1 m	◐ 1 m	1
L2 · courtyard NE corner	2.1 m	● 1.56	● 6.3 / 2.7 m	◐ two family rooms	●	●	○ 0 · 1.4 m	● 4.9 m	0.85
L3 · west band	2.2 m	● 1.53	○ 1.8 / 9 m	○ an en-suite bath	●	◐	● 3 · 0 m	○ 0 m	0.95
L4 · kitchen, east band	5.7 m	● 1.51	○ 3 / 1.9 m	○ suite bath, service route	○	○	○ 0 · 0.6 m	○ 0 m	1.15

● good · ◐ acceptable / with a price · ○ poor or blocking. Breeze: the relative courtyard flow (today = 1.00). Intake: the distance to the nearest plumbing vent / to the condensers. Gate · street, courtyard: the visible height of the tower (m).

7 · Recommendation: keep the tower at the north-east cut corner (L1)

- **L1 is the best all round:** it takes only service space (the maid bath and bedroom 1 each lose ≈ 0.8 m², each gains a store or a cupboard), keeps every family and guest room whole, gives the cleanest HRV intake (8.3 m from the nearest vent, 4.6 m from the condensers), leaves the roof's zones in place and is **the one seen as the symbol of the house from outside**: 3.3 m of the tower above the coping from the gate, 4.1 m from the street, its crown from the courtyard (1 m).
- **Its one weakness, the long breeze route (10.6 m), is mostly the throat, not the length:** with the channel kept at its full 0.9 × 0.2 under the re-planned maid bath (the WC and the shower drains off its line, in the interior study), the flow rises from 0.99 to **1.44** × today's, close to the 1.56 of the courtyard corner.
- **L2, the courtyard corner, is the strong alternative** and the true Yazd position (the badgir over the hall, facing the courtyard): the shortest route (2.1 m) straight onto the seating, the best structure (on W3 and the courtyard wall, no plenum), the cheapest, and 4.9 m of the tower rising over the courtyard's corner; **but** it takes ≈ 1.4 m² as a nib from the women's majlis and from the family living, and it is **not seen from the gate** (0 m), only from the street (1.4 m). Choose it if the tower is to be the symbol of the courtyard rather than of the house's face.
- **Not recommended:** L3 (it costs bedroom 2's en-suite its service wall and sits on the west stack's vent) and L4 (it closes the kitchen's route to the service door, stands in the master suite's bath and lands in the condensers' zone).
- **So:** the tower of pages 1–5 at L1, with the channel at its full width; the plan, the elevations and the gate render of pages 1–4 stand, and no new renders are needed for the position.

ملخص الموقع

قورنت أربعة مواقع للملقف: الركن الشمالي الشرقي عند جدار القطع (الحالي)، وركن الفناء الشمالي الشرقي داخل مجلس النساء، والشريط الغربي بجوار شفت الخدمات، وفوق المطبخ في الشريط الشرقي.

التوصية: إبقاء البرج في الركن الشمالي الشرقي: لا يأخذ إلا من غرف الخدمة، ومدخل هواء التهوية فيه الأنظف، ويُرى من البوابة والشارع رمزا للبيت. ضعفه الوحيد طول مجرى النسيم (≈ 10.6 م)، وأكثر خسارته من الاختناق تحت حمام الخادمة؛ فإذا بقي المجرى بعرضه الكامل ارتفع التدفق إلى ما يقارب موقع الفناء.

البديل القوي: ركن الفناء (موقع الملقف اليزدي الأصيل): أقصر مجرى، وأفضل إنشائيا وأرخص، ويظهر البرج فوق ركن الفناء؛ لكنه يأخذ نحو 1.4 م² من مجلس النساء ومن جلسة العائلة ولا يُرى من البوابة. لا يُنصح بالشريط الغربي ولا بالمطبخ.